

1 Classical Wave Equation with Pinned Ends

This section describes the standard **one-dimensional wave equation** defined over the domain $x \in [0, L]$ and time $t > 0$. This configuration typically models a vibrating string where the ends are fixed.

Governing Equation (Strong Form):

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (1)$$

where $u(x, t)$ is the displacement, and c is the constant wave speed.

Boundary Conditions (Dirichlet/Essential): The boundary conditions are the classic **pinned ends** or **Dirichlet** conditions:

$$\begin{aligned} u(0, t) &= 0 \\ u(L, t) &= 0 \end{aligned}$$

Initial Conditions: The initial conditions define the state of the wave at $t = 0$:

- **Initial Displacement:**

$$u(x, 0) = f(x)$$

- **Initial Velocity:**

$$\frac{\partial u}{\partial t}(x, 0) = g(x)$$

2 Wave Equation with First-Order Absorbing Boundary Conditions (ABCs)

This configuration uses non-reflecting boundaries, which are critical in numerical methods (like Finite Difference or Finite Element) to prevent artificial wave reflections from the domain edges. These are the **first-order Absorbing Boundary Conditions (ABCs)**.

Governing Equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad (2)$$

2.0.1 Decomposition of the Wave Operator

The strong form of the wave equation can be factored into two first-order differential operators, each corresponding to a distinct wave direction. This

decomposition is fundamental to understanding the **one-way wave equation** and the ABC derivation.

$$\left(\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} \right) u(x, t) = \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) u(x, t) = 0 \quad (3)$$

The term $\left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) u = 0$ corresponds to a wave moving to the **right**, while $\left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) u = 0$ corresponds to a wave moving to the **left**.

First-Order Absorbing Boundary Conditions (ABCs): To enforce a non-reflecting boundary, we apply the one-way wave operator that corresponds to the **outgoing** wave direction.

- **At $x = 0$ (Outgoing Wave Moving Left $\leftarrow$):** We assume the field u is composed only of the outgoing wave component, forcing the right-moving component to be zero:

$$\frac{\partial u}{\partial t}(0, t) + c \frac{\partial u}{\partial x}(0, t) = 0 \quad \implies \quad \frac{\partial u}{\partial x}(0, t) + \frac{1}{c} \frac{\partial u}{\partial t}(0, t) = 0$$

- **At $x = L$ (Outgoing Wave Moving Right $\rightarrow$):** We assume the field u is composed only of the outgoing wave component, forcing the left-moving component to be zero:

$$\frac{\partial u}{\partial t}(L, t) - c \frac{\partial u}{\partial x}(L, t) = 0 \quad \implies \quad \frac{\partial u}{\partial x}(L, t) - \frac{1}{c} \frac{\partial u}{\partial t}(L, t) = 0$$